

SensBlue Monarch — How it compares

Feature comparison against similar cellular IoT
boards

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1. Purpose and scope

Products were selected based on functional overlap with the SensBlue Monarch: cellular connectivity (LTE-M / NB-IoT / LTE Cat-1), user-programmable, and at least one industrial or sensing capability. Boards without cellular (LoRa-only, Wi-Fi-only) were excluded, as were pure modem modules requiring an external MCU.

This document is a factual, feature-focused comparison of the SensBlue Monarch against similar boards from the maker, prototyping and light-industrial segments. Its purpose is to help engineers make an informed selection when specifying a cellular IoT edge device.

2. Categories of comparable boards

Tier A — Chinese maker boards

ESP32-based boards paired with a cellular modem module (e.g., SIMCom SIM7000, SIM7080, SIM7670 or Air720 families). Typically sold on AliExpress, Amazon and specialist stores; aimed primarily at makers and educational projects. The cellular module carries its own certifications; the finished board is generally not separately certified for the EU. Support model is community-driven (forums, vendor GitHub repositories).

Tier B — Western maker / prototyping boards

Cellular development kits from Particle, Blues Wireless, Nordic Semiconductor and Arduino. Higher polish, integrated cloud services or secure elements, and OTA update tooling, typically paired with a subscription or usage-based business model. Regulatory scope varies — some products target only the North American market, others are certified globally. Consult each vendor's certifications page before selection.

Tier C — Industrial cellular gateways

Industrial-grade cellular routers and gateways from Teltonika, Robustel, Advantech, Sierra Wireless and equivalents. Linux/OpenWrt-based, DIN-rail mountable, focused on connectivity (WAN failover, VPN, remote management). Programmable through vendor SDKs (RutOS, custom Linux). On-board environmental sensing, isolated analog I/O and battery management are not usual for this device class.

3. Detailed comparison

The table below summarises the most relevant competitors across the three tiers. Column definitions follow the table.

Product	Origin	MCU	Cellular	GNSS	Sensors	Iso DI/DO	Analog	RS485	Power	EU cert.	
LILYGO T-SIM7000G	China	ESP32-WROVER-B	LTE-M / NB-IoT (SIM7000G)	GPS	—	—	—	—	18650 + solar	Modem only	Yes
Waveshare ESP32-S3 SIM7080G	China	ESP32-S3	LTE-M / NB-IoT (SIM7080G)	GNSS	—	—	—	—	USB only	Modem only	Yes
Waveshare ESP32-S3 SIM7670G	China	ESP32-S3	LTE Cat-1 (SIM7670G)	GNSS	—	—	—	—	USB only	Modem only	Yes
LILYGO T-A7670G-H	China	ESP32	LTE Cat-1 (A7670G)	GNSS	—	—	—	—	18650	Modem only	Yes
Particle Boron LTE-M	USA	Nordic nRF52840	LTE-M Cat.M1 (u-blox SARA-R510)	—	—	—	—	—	LiPo charge	No (NorAm)	Partial
Blues Notecard Cellular	USA	STM32 + modem	LTE-M / NB-IoT or Cat-1 bis	GNSS	—	—	—	—	external	PTCRB module	Yes
Nordic Thingy:91	Norway	nRF9160 SiP	LTE-M / NB-IoT	GPS	T/H/P/gas/motion	—	—	—	LiPo	Yes (global)	No (C SDK)
Arduino MKR NB 1500	Italy	SAMD21 + SARA-R410	NB-IoT / LTE-M	—	—	—	—	—	external	Yes	Yes
Teltonika TRB142	LT	ARM Cortex-A7	LTE Cat 1	—	—	—	—	RS232	external	CE/UKCA/FCC	No (RutOS)
Teltonika TRB246	LT	ARM Cortex-A7	LTE Cat 4 / dual SIM	—	—	2x DI/DO	—	RS232/485	external	CE	No (RutOS)
Robustel R1511	CN/DE	ARM Cortex-A7	LTE Cat 4	optional	—	—	—	1x RS232	external	CE	No (Linux)
ATRONIA SensBlue Monarch	PT	ESP32-WROOM-32E-N8	LTE-M / NB-IoT / EGPRS (BG95-M3)	GPS/GLONASS/BeiDou/Galileo	BME280 T/H/P	2x isolated	2x 0-10V / 4-20mA	1x 15kV ESD	USB+battery+solar (BQ25176)	CE (planned)	Yes

Legend: “—” = feature not present.

Column definitions

- Cellular — radio access technology and modem chip.

- GNSS — satellite positioning system present on the board (not just addressable through an external module).
- Sensors — environmental sensors integrated on the board itself.
- Iso DI/DO — number of galvanically isolated digital inputs / outputs suitable for 24 V field wiring.
- Analog — configurable analog inputs (0-10 V / 4-20 mA) for industrial sensors.
- RS485 — half-duplex RS485 transceiver with ESD protection; not present is different from a bare UART.
- Power — supply options and battery management. “Solar” indicates a properly-designed charge controller, not just an unregulated input.
- EU cert. — regulatory certification status for the finished product in the European Union. “Modem only” means the modem module is certified but the assembly is not.
- Arduino — programmable directly from the Arduino IDE without workarounds. “Partial” indicates Arduino-compatible tooling with limitations (e.g., custom cloud dependencies).

4. Data sources and notes

Specifications were collected from the following public sources in September 2026:

- Particle: store.particle.io, docs.particle.io
- Blues Wireless: shop.blues.com, blues.com/products
- Nordic Semiconductor: nordicsemi.com, distributors (Digi-Key, Rutronik)
- LILYGO: lilygo.cc, opencircuit.shop, AliExpress LILYGO official store
- Waveshare: waveshare.com, esp32s.com, Amazon
- Teltonika Networks: teltonika-networks.com and EU distributors (Getic, Sectron, WiFi-Stock)
- Advantech: advantech.com (WISE-2410 was found to be a LoRaWAN vibration sensor, not comparable and therefore excluded)
- Robustel: robustel.com and EU distributors

Where a product had multiple regional variants (e.g. Particle Boron NorAm vs global), the highest-availability variant in the EU market was used. Certifications marked "planned" or "in progress" reflect the product owner's own communications and were not independently verified.

Product lines and feature sets change quickly in this segment. Verify with each vendor before selection. For up-to-date information on the SensBlue Monarch itself see the datasheet and the project repository.